

WASP-52b

R-1046

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_161006 · created 2026-07-30T09:16:54+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457667.6942 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.062
Transit depth (Rp/Rs)^2	3.53 +/- 2.34 [%]
Orbital Inclination (inc)	82.46 +/- 2.76
Airmass coefficient 1 (a1)	1.89 +/- 1.25
Airmass coefficient 2 (a2)	-0.47 +/- 0.46
Scatter in the residuals of the lightcurve fit is	68.87 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	6.25
Transit Duration (day)	0.043 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.188 ± 0.062 vs 0.1646 (deviation 0.36σ)
Duration fit vs published	0.043 d vs 0.0758 d (deviation 0.99σ)
Mid-time O–C	-4.6 min
Depth SNR	2.9
Residual scatter	68.87 %

Light curve

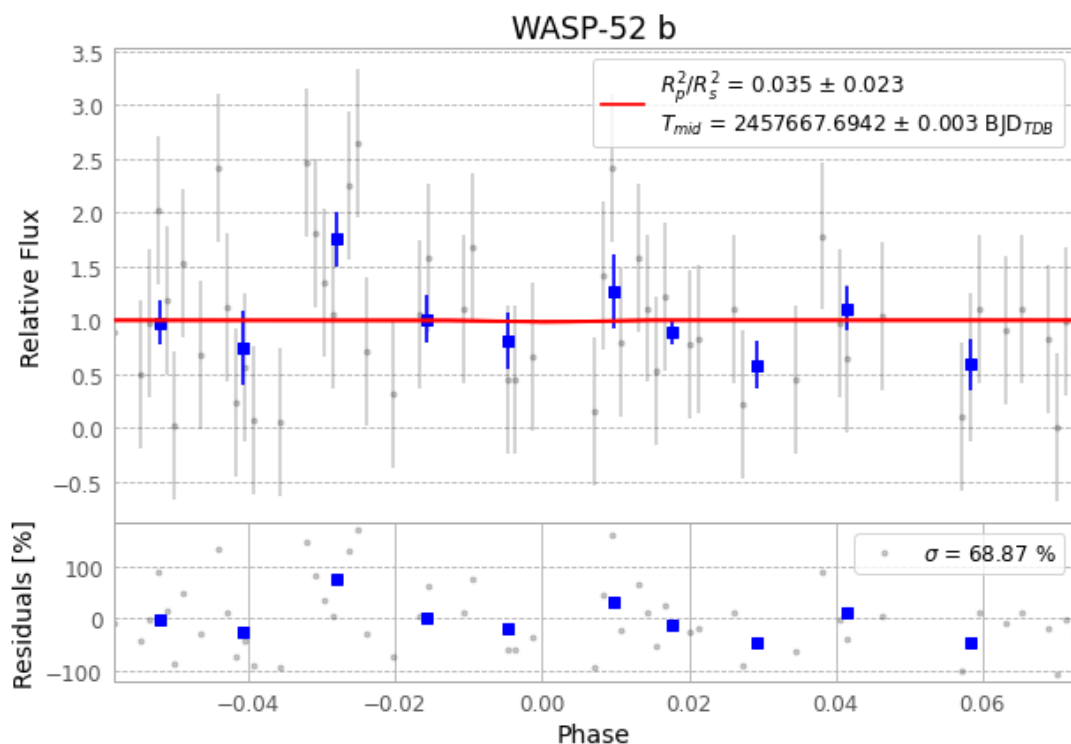


Figure 1 — FinalLightCurve_WASP-52 b_2016-10-05.png (SHA-256 437a209b0de8a778...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 131], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fab4fe6b76faa5be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

112 FITS frames (74 MB), WASP-52161006020312.FITS → WASP-52161006073612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-161006.log (d4b70b9cb11b...), FinalLightCurve_WASP-52 b_2016-10-05.png (437a209b0de8...), FinalLightCurve_WASP-52 b_2016-10-05.csv (726d3b9547d4...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 09:16Z.